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Tue 23 Jun 2026 07:23:18 - sadmlib_std_demo.sh v3.30 - Library v4.90.3
Desc: Demonstrate functions & variables available to developers using SADMIN Tools
bilbo - User: root - Umask : 0022 - Arch: x86_64
Fedora Linux v43 - Kernel 7.0.12-101.fc43.x86_64 - SADMIN(C): /opt/sadmin
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User Var. that affect SADMIN behavior	Description	This System Result
[001] \$SADM_VER	Script Version Number	: 3.30
[002] \$SADM_PN	Script Name	: sadmlib_std_demo.sh
[003] \$SADM_INST	Script Name Without Extension	: sadmlib_std_demo
[004] \$SADM_USERNAME	Current User Name	: root
[005] \$SADM_TPID	Current Process ID	: 348066
[006] \$SADM_MULTIPLE_EXEC	Allow running multiple copy [Y/N]	: Y
[007] \$SADM_USE_RCH	Generate entry in '*.rch' file	: N
[008] \$SADM_LOG_TYPE	Set Output to [S]creen [L]og [B]oth	: B
[009] \$SADM_LOG_APPEND	Y=Append to Log, N=Create new log	: N
[010] \$SADM_LOG_HEADER	Generate Header in log [Y,N]	: Y
[011] \$SADM_LOG_FOOTER	Generate Footer in log [Y/N]	: Y
[012] \$SADM_EXIT_CODE	Current value of script exit code	: 0
[013] \$SADM_ROOT_ONLY	Only run by 'root' user [Y/N]	: N
[014] \$SADM_SERVER_ONLY	Only run on SADMIN server [Y/N]	: N

Calling Functions	Description	This System Result
[001] \$(sadm_get_release)	SADMIN Release Number (XX.XX)	: 2.1.2
[002] \$(sadm_get_ostype)	OS Type (Uppercase,LINUX,AIX,DARWIN)	: LINUX
[003] \$(sadm_get_osversion)	Return O/S Version (Ex: 7.2, 6.5)	: 43
[004] \$(sadm_get_osmajorversion)	Return O/S Major Version (Ex 7, 6)	: 43
[005] \$(sadm_get_osminorversion)	Return O/S Minor Version (Ex 2, 3)	:
[006] \$(sadm_get_osname)	O/S Name (REDHAT,CENTOS,UBUNTU,...)	: FEDORA
[007] \$(sadm_get_oscodename)	O/S Project Code Name	: n/a
[008] \$(sadm_get_kernel_version)	O/S Running Kernel Version	: 7.0.12-101.fc43.x86_64
[009] \$(sadm_get_kernel_bitmode)	O/S Kernel Bit Mode (32 or 64)	: 64
[010] \$(sadm_get_hostname)	Current Host Name	: bilbo
[011] \$(sadm_get_host_ip)	Current Host IP Address	: 192.168.1.23
[012] \$(sadm_get_domainname)	Current Host Domain Name	: maison.ca
[013] \$(sadm_get_fqdn)	Fully Qualified Domain Host Name	: bilbo
[014] \$(sadm_server_serial)	Server serial number (Ex: 4S7GYF1)	: E537F839
[015] \$(sadm_get_epoch_time)	Get Current Epoch Time	: 1782213799
[016] \$(sadm_epoch_to_date 1782213799)	Convert epoch time to date	: 2026.06.23 07:23:19
	WDATE='2026.06.23 07:23:19'	
[017] \$(sadm_date_to_epoch "\$WDATE")	Convert Date to epoch time	: 1782213799

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DATE1='2016.01.30 10:00:44'
DATE2='2016.01.30 10:00:03'
[018] $(sadm_elapse "$DATE1" "$DATE2") Elapse Time between two timestamps : 00:00:41
[019] $(sadm_get_packagetype) Get package type (rpm,deb,aix,dmg) : rpm
[020] $(sadm_server_arch) System Architecture : x86_64
[021] sadm_lock_system "hostname" Lock the specified hostname : 0=Lock 1=Error not lock
[022] sadm_lock_status "hostname" Check if specified hostname is lock : 0=Not Lock - 1=System Locked
[023] sadm_unlock_system "hostname" Unlock the specified hostname : 0=Unlocked 1=Error could not unlock
[024] sadm_writelog "message" Depreciated use 'sadm_write_log' : message
[025] sadm_write_log "message" Write message to Screen, Log or Both: message
[026] sadm_write_err "message" Write mess to error log & to std log: message
[027] sadm_sendmail(mail,sub,body,att) sadm_sendmail(email,'subject',body',file1,...): 0=Success 1=Error
[028] sadm_show_version Cmdline -v option show script info. :
[029] sadm_sleep "60" "15" Sleep 60sec. at interval of 15sec. : 60...45...30...15...0
[030] sadm_trimfile "filename" 35 Keep the last 35 lines of filename. : 0=Success 1=Error

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FUNCTIONS AVAIL. ONLY IN BASH LIBRARY Description This System Result
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[001] $(sadm_server_type) Host is Physical or Virtual (P/V) : P
[002] $(sadm_server_model) Server model (Ex: HP ProLiant DL580): ThinkPad T470 W10DG
[003] $(sadm_server_memory) Server total memory in MB (Ex: 3790): 31963
[004] $(sadm_server_hardware_bitmode) CPU Hardware capable of 32/64 bits : 64
[005] $(sadm_server_nb_logical_cpu) Number of Logical CPU on system : 4
[006] $(sadm_server_nb_cpu) Number of Physical CPU on system : 1
[007] $(sadm_server_nb_socket) Number of socket on system : 1
[008] $(sadm_server_core_per_socket) Number of Core per Socket : 2
[009] $(sadm_server_thread_per_core) Number of Thread per Core : 2
[010] $(sadm_server_cpu_speed) Server CPU Speed in MHz : 3000
[011] $(sadm_server_disks) Disks list(MB) (DISKNAME|SIZE,...) : sdb|1024000,nvme0n1|524288
[012] $(sadm_server_vg) VG list(MB) (VGNAME|SIZE|USED|FREE) : No VG
[013] $(sadm_server_ips) Network IP (Name|IP|Netmask|MAC) : enp0s31f6|192.168.1.23|255.255.255.0|54:e1:ad:75:8e:0e,virbr0|192.168.124.1|255.255.255.0|52:54:00:0c:03:dd
[014] sadm_toupper string Return string uppercase : STRING
[015] sadm_tolower STRING Return string lowercase : string
[016] sadm_capitalize STRING Return string capitalize : String
[017] sadm_ask "Question" Show message & wait for [Y/y/N/n] : Question [Y/N] ? - (0=N 1=Y)

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Overview of the 'sa.start' and 'sa.stop' functions in Bash Library.

=====Bash Library Function

'start()' Bash Library Function

The 'start()' function basically initialize the SADMIN environment.\n
When you call 'sadm_start()', it will only come back to caller if everything went OK.
Otherwise it will advise the user of the error and exit(1).

Here is a summary of the different things it does :

- 1) Make sure all \$SADMIN directories & sub-dir. exist and have proper permissions.
- 2) If \$SADM_LOG_APPEND='N', create new log & error log, else append to actual log.
- 3) If \$SADM_LOG_HEADER='Y', write the log header.
- 4) if \$SADM_ROOT_ONLY='Y', and current user is not root, show error message and exit(1).
- 5) If \$SADM_SERVER_ONLY='Y', and not on the SADMIN server, show error message and exit(1).
- 6) If \$SADM_SADMGRP_ONLY='Y', and the user is not part of the SADM_GROUP (unless 'root'), show error message and exit(1).
- 7) If \$SADM_USE_DB='Y' but not on SADMIN Server, show error message and exit(1).
- 8) If \$SADM_USE_RCH='Y', write starting time to the RCH file (With a code 2=Running).
- 9) If system is lock (and running on the SADMIN), issue message and exit(1)
- 10) If PID file '\$SADM_PID_FILE' exist and '\$SADM_MULTIPLE_EXEC='Y'', continue normal execution
If PID file exist and execution time (sec) is less than the '\$SADM_PID_TIMEOUT', show error message and exit(1).
If PID file exist and execution time (sec) exceed the '\$SADM_PID_TIMEOUT' a new '\$SADM_PID_FILE' is created and execution is resume.

*** If any unrecoverable error occurs while executing this function the program is aborted exit(1). ***

'stop(exit_code)' Bash Library Function

- exit_code (0=Success, 1=Error): Variable defined in SADMIN section.

This should be the one of the last function called at the end of your program.

What this function does:

- 1) Calculate execution Time.
- 2) If N='Y', update the rch file (End Time & Elapse Time ...).
- 3) Validate the alert group,
- 4) If Y='Y', write the log footer.
- 5) If the error log file is empty, then delete it.
- 6) Delete the PID file (/opt/sadmin/tmp/sadmlib_std_demo.pid) of the program.
- 7) Close log and error log files.
- 8) If 500 is not zero, trim the log according to user choice in '500'.
- 9) If 30 is not zero, trim the 'rch' according to user choice in 'SADM_MAX_RCLINE'.
- 10) Set permission and owner/group to log and rch files.
- 11) If on the SADMIN server, then rch and log are immediatly web central directory.

SADMIN CONFIG FILE VARIABLES	Description	This System Result
[001] \$SADM_SERVER	SADMIN SERVER NAME (FQDN)	: sherlock.maison.ca
[002] \$SADM_HOST_TYPE	SADMIN [C]lient or [S]erver	: C
[003] \$SADM_MAIL_ADDR	SADMIN Administrator Default Email	: brucetalbot95@gmail.com
[004] \$SADM_ALERT_TYPE	0=NoMail 1=OnError 2=OnSuccess 3=All:	1

[005]	\$SADM_ALERT_GROUP	Default Alert Group	: default
[006]	\$SADM_ALERT_REPEAT	Seconds to wait before repeat alert	: 0
[007]	\$SADM_DAYS_HISTORY	Days to keep alert in History file	: 14
[008]	\$SADM_MAX_ARC_LINE	Max lines to keep in alert Archive	: 1000
[009]	\$SADM_TEXTBELT_KEY	TextBelt.com API Key	: *Hidden*
[010]	\$SADM_TEXTBELT_URL	TextBelt.com API URL	: https://textbelt.com/text
[011]	\$SADM_CIE_NAME	Your Company Name	: TheBatcave
[012]	\$SADM_DOMAIN	Server Creation Default Domain	: maison.ca
[013]	\$SADM_USER	SADMIN User Name	: sadmin
[014]	\$SADM_GROUP	SADMIN Group Name	: sadmin
[015]	\$SADM_PWD_RANDOM	Auto generation of 'sadmin' password:	Y
[016]	\$SADM_WWW_USER	User that Run Apache Web Server	: apache
[017]	\$SADM_WWW_GROUP	Group that Run Apache Web Server	: apache
[018]	\$SADM_DBNAME	SADMIN Database Name	: sadmin
[019]	\$SADM_DBHOST	SADMIN Database Host	: localhost
[020]	\$SADM_DBPORT	SADMIN Database Host TCP Port	: 3306
[021]	\$SADM_RW_DBUSER	SADMIN Database Read/Write User	: sadmin
[022]	\$SADM_RW_DBPWD	SADMIN Database Read/Write User Pwd	: *Hidden*
[023]	\$SADM_RO_DBUSER	SADMIN Database Read Only User	: squery
[024]	\$SADM_RO_DBPWD	SADMIN Database Read Only User Pwd	: *Hidden*
[025]	\$SADM_SMTP_SERVER	Your Internet SMTP Server Name	: smtp.gmail.com
[026]	\$SADM_SMTP_PORT	Your Internet SMTP Server Port	: 587
[027]	\$SADM_SMTP_SENDER	Your Internet Sender Email	: brucetalbot95@gmail.com
[028]	\$SADM_GMPW	Your Internet Sender Password	: *Hidden*
[029]	\$SADM_SSH_PORT	SSH Port to communicate with client	: 32
[030]	\$SADM_MONITOR_UPDATE_INTERVAL	Monitor page update interval in sec.	: 60
[031]	\$SADM_MONITOR_RECENT_COUNT	Monitor page nb. recent scripts	: 5
[032]	\$SADM_MONITOR_RECENT_EXCLUDE	Scripts excluded from SysMon Recent	: sadm_nmon_watcher
[033]	\$SADM_NMON_KEEPPAYS	Nb. of days to keep nmon perf. file	: 10
[034]	\$SADM_RCH_KEEPPAYS	Nb. days to keep unmodified rch file:	35
[035]	\$SADM_LOG_KEEPPAYS	Nb. days to keep unmodified log file:	35
[036]	\$SADM_MAX_RCLINE	Trim rch file to this max. of lines	: 30
[037]	\$SADM_MAX_LOGLINE	Trim log to this maximum of lines	: 500
[038]	\$SADM_NETWORK1	Network/Netmask 1 inv. IP/Name/Mac	: 192.168.1.0/24
[039]	\$SADM_NETWORK2	Network/Netmask 2 inv. IP/Name/Mac	: None
[040]	\$SADM_NETWORK3	Network/Netmask 3 inv. IP/Name/Mac	: None
[041]	\$SADM_NETWORK4	Network/Netmask 4 inv. IP/Name/Mac	: None
[042]	\$SADM_NETWORK5	Network/Netmask 5 inv. IP/Name/Mac	: None
[043]	\$SADM_REAR_NFS_SERVER	Rear NFS Server IP or Name	: lestrade.maison.ca
[044]	\$SADM_REAR_NFS_MOUNT_POINT	Rear NFS Mount Point	: /backup/backup_rear
[045]	\$SADM_REAR_BACKUP_TO_KEEP	Rear NFS Backup - Nb. to keep	: 2
[046]	\$SADM_REAR_BACKUP_DIF	Alert if size differ more than %	: 25%
[047]	\$SADM_REAR_BACKUP_INTERVAL	Rear alert if Backup older than	: 14 days
[048]	\$SADM_BACKUP_NFS_SERVER	NFS Backup IP or Server Name	: batnas.maison.ca
[049]	\$SADM_BACKUP_NFS_MOUNT_POINT	NFS Backup Mount Point	: /volumel/backup/backup_sadmin
[050]	\$SADM_BACKUP_DIF	Alert if size differ more than %	: 40%
[051]	\$SADM_BACKUP_INTERVAL	Alert if Backup older than X days	: 1 days
[052]	\$SADM_DAILY_BACKUP_TO_KEEP	Nb. of Daily Backup to keep	: 2

[053]	\$SADM_WEEKLY_BACKUP_TO_KEEP	Nb. of Weekly Backup to keep	: 1
[054]	\$SADM_MONTHLY_BACKUP_TO_KEEP	Nb. of Monthly Backup to keep	: 1
[055]	\$SADM_YEARLY_BACKUP_TO_KEEP	Nb. of Yearly Backup to keep	: 1
[056]	\$SADM_WEEKLY_BACKUP_DAY	Weekly Backup Day (1=Mon,...,7=Sun)	: 5
[057]	\$SADM_MONTHLY_BACKUP_DATE	Monthly Backup Date (1-28)	: 1
[058]	\$SADM_YEARLY_BACKUP_MONTH	Month to take Yearly Backup (1-12)	: 12
[059]	\$SADM_YEARLY_BACKUP_DATE	Date to do Yearly Backup(1-DayInMth):	31
[060]	\$SADM_PID_TIMEOUT	PID File Time To Live default in sec:	7200
[061]	\$SADM_LOCK_TIMEOUT	Maximum of sec. a system can be lock:	3600
[062]	\$SADM_VM_EXPORT_NFS_SERVER	NFS Export Server	: batnas.maison.ca
[063]	\$SADM_VM_EXPORT_MOUNT_POINT	NFS Export Mount Point	: /volumel/backup/backup_vm/virtualbox_exports
[064]	\$SADM_VM_EXPORT_TO_KEEP	Nb. of export to keep	: 2
[065]	\$SADM_VM_EXPORT_INTERVAL	Days without export before alert	: 21
[066]	\$SADM_VM_EXPORT_ALERT	Issue an alert if interval reached	: Y
[067]	\$SADM_VM_USER	User part of 'vboxusers' user group	: jacques
[068]	\$SADM_VM_STOP_TIMEOUT	Max. seconds given for acpi shutdown:	60
[069]	\$SADM_VM_START_INTERVAL	Sec. to sleep between each VM start	: 30

Directories	Var. Avail.	Description	This System Result
[001]	\$SADM_BASE_DIR	SADMIN Root Directory	: /opt/sadmin
[002]	\$SADM_BIN_DIR	SADMIN Scripts Directory	: /opt/sadmin/bin
[003]	\$SADM_TMP_DIR	SADMIN Temporary file(s) Directory	: /opt/sadmin/tmp
[004]	\$SADM_LIB_DIR	SADMIN Shell & Python Library Dir.	: /opt/sadmin/lib
[005]	\$SADM_LOG_DIR	SADMIN Script Log Directory	: /opt/sadmin/log
[006]	\$SADM_CFG_DIR	SADMIN Configuration Directory	: /opt/sadmin/cfg
[007]	\$SADM_SYS_DIR	Server Startup/Shutdown Script Dir.	: /opt/sadmin/sys
[008]	\$SADM_DOC_DIR	SADMIN Documentation Directory	: /opt/sadmin/doc
[009]	\$SADM_PKG_DIR	SADMIN Packages Directory	: /opt/sadmin/pkg
[010]	\$SADM_DAT_DIR	Server Data Directory	: /opt/sadmin/dat
[011]	\$SADM_NMON_DIR	Server NMON - Data Collected Dir.	: /opt/sadmin/dat/nmon
[012]	\$SADM_DR_DIR	Server Disaster Recovery Info Dir.	: /opt/sadmin/dat/dr
[013]	\$SADM_RCH_DIR	Server Return Code History Dir.	: /opt/sadmin/dat/rch
[014]	\$SADM_NET_DIR	Server Network Info Dir.	: /opt/sadmin/dat/net
[015]	\$SADM_RPT_DIR	SYStem MONitor Report Directory	: /opt/sadmin/dat/rpt
[016]	\$SADM_DBB_DIR	Database Backup Directory	: /opt/sadmin/dat/dbb
[017]	\$SADM_SETUP_DIR	SADMIN Installation/Update Dir.	: /opt/sadmin/setup
[018]	\$SADM_USR_DIR	User/System specific directory	: /opt/sadmin/usr
[019]	\$SADM_UBIN_DIR	User/System specific bin/script Dir.	: /opt/sadmin/usr/bin
[020]	\$SADM_ULIB_DIR	User/System specific library Dir.	: /opt/sadmin/usr/lib
[021]	\$SADM_UDOC_DIR	User/System specific documentation	: /opt/sadmin/usr/doc
[022]	\$SADM_UMON_DIR	User/System specific SysMon Scripts	: /opt/sadmin/usr/mon
[023]	\$SADM_WWW_DIR	SADMIN Web Site Root Directory	: /opt/sadmin/www
[024]	\$SADM_WWW_DOC_DIR	SADMIN Web Documentation Dir.	: /opt/sadmin/www/doc
[025]	\$SADM_WWW_DAT_DIR	SADMIN Web Site Systems Data Dir.	: /opt/sadmin/www/dat
[026]	\$SADM_WWW_LIB_DIR	SADMIN Web Site PHP Library Dir.	: /opt/sadmin/www/lib
[027]	\$SADM_WWW_TMP_DIR	SADMIN Web Temp Working Directory	: /opt/sadmin/www/tmp

[028] \$SADM_WWW_PERF_DIR SADMIN Web Performance Graph Dir. : /opt/sadmin/www/tmp/perf

SADMIN FILES VARIABLES AVAIL.	Description	This System Result
[001] \$SADM_PID_FILE	Current script PID file	: /opt/sadmin/tmp/sadmlib_std_demo.pid
[002] \$SADM_CFG_FILE	SADMIN Configuration File	: /opt/sadmin/cfg/sadmin.cfg
[003] \$SADM_CFG_HIDDEN	SADMIN Initial Configuration File	: /opt/sadmin/cfg/.sadmin.cfg
[004] \$SADM_ALERT_FILE	SADMIN Alert Group File	: /opt/sadmin/cfg/alert_group.cfg
[005] \$SADM_ALERT_INIT	SADMIN Initial Alert Group File	: /opt/sadmin/cfg/.alert_group.cfg
[006] \$SADM_ALERT_HIST	SADMIN Alert History File	: /opt/sadmin/cfg/alert_history.txt
[007] \$SADM_ALERT_HINI	SADMIN Initial Alert History File	: /opt/sadmin/cfg/.alert_history.txt
[008] \$SADM_TMP_FILE1	User usable Temp Work File 1	: /opt/sadmin/tmp/sadmlib_std_demo1_4SG
[009] \$SADM_TMP_FILE2	User usable Temp Work File 2	: /opt/sadmin/tmp/sadmlib_std_demo2_bSZ
[010] \$SADM_TMP_FILE3	User usable Temp Work File 3	: /opt/sadmin/tmp/sadmlib_std_demo3_4RE
[011] \$SADM_LOG	Script Log File	: /opt/sadmin/log/bilbo_sadmlib_std_demo.log
[012] \$SADM_RCH_FILE	Script Return Code History File	: /opt/sadmin/dat/rch/bilbo_sadmlib_std_demo.rc
h		
[013] \$DBPASSFILE	SADMIN Database User Password File	: *Hidden*
[014] \$SADM_RPT_FILE	System Monitor Report File	: /opt/sadmin/dat/rpt/bilbo_sadmlib_std_demo.rp
t		
[015] \$SADM_BACKUP_LIST	Backup List File Name	: /opt/sadmin/cfg/backup_list.txt
[016] \$SADM_BACKUP_LIST_INIT	Initial Backup List (Template)	: /opt/sadmin/cfg/.backup_list.txt
[017] \$SADM_BACKUP_EXCLUDE	Backup Exclude List File Name	: /opt/sadmin/cfg/backup_exclude.txt
[018] \$SADM_BACKUP_EXCLUDE_INIT	Initial Backup Exclude (Template)	: /opt/sadmin/cfg/.backup_exclude.txt

COMMAND PATH USE BY SADMIN STD. LIBR.	Description	This System Result
[001] \$SADM_DMIDECODE	'dmidecode' use to get model & type	: /usr/sbin/dmidecode
[002] \$SADM_BC	'bc' use to do some Math.	: /usr/bin/bc
[003] \$SADM_FDISK	'fdisk' use to get partition info	: /usr/sbin/fdisk
[004] \$SADM_WHICH	'which' use to get command path	: /usr/bin/which
[005] \$SADM_PERL	'perl' epoch time Math on AIX	: /usr/bin/perl
[006] \$SADM_MUTT	'mutt' use to send email	: /usr/bin/mutt
[007] \$SADM_CURL	'curl' Use to send alert to Slack	: /usr/bin/curl
[008] \$SADM_LSCPU	'lscpu' To get Socket/Thread info	: /usr/bin/lscpu
[009] \$SADM_NMON	'nmon' Collect Performance Stat.	: /usr/bin/nmon
[010] \$SADM_PARTED	'parted' use to get disk real size	: /usr/sbin/parted
[011] \$SADM_ETHTOOL	'ethtool' use to get system IP	: /usr/sbin/ethtool
[012] \$SADM_SSH	'ssh' use to SSH on SADMIN client	: /usr/bin/ssh
[013] \$SADM_SSH_CMD	'ssh' SSH cmd to Connect to client	: /usr/bin/ssh -qnp 32
[014] \$SADM_RRDTOOL	'rrdtool' binary location	: /usr/bin/rrdtool
[015] \$SADM_LSB_RELEASE	'lsb_release' cmd (get dist. info)	: /usr/bin/lsb_release
[016] \$SADM_INXI	'inxi' binary location	: /usr/bin/inxi
[017] \$SADM_MYSQL	'mysql' binary location	: /usr/bin/mysql
[018] \$SADM_SED	'sed' binary location	: /usr/bin/sed

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Script exit code is 0 (Success) and execution time was 00:00:02.
Script is set to only send an alert when it terminate with error.
Script succeeded, no alert will be send ($SADM_ALERT_TYPE=1).
New log created '$SADMIN/log/bilbo_sadmlib_std_demo.log'.
End of sadmlib_std_demo.sh - Tue 23 Jun 2026 07:23:20 AM EDT
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